Momentum analysis: put it together

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use a diagram or calculation to compare a deflection option. Explain where your prediction stops working.

SUCCESS CRITERIA - I can...

1. Compute the required change before evaluating an option.
2. Keep impulse, velocity change and displacement distinct.
3. Give the numerical result and state what the calculation treats as true.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “deflection” and “lead time” to explain your answer. Sentence starters are optional.

deflection: a change in the direction or path of motion

lead time: time available before the modeled event

system: the object or objects chosen for analysis

01 Warm-up: bring back an earlier idea

A system’s total momentum is **15 kg·m/s** *west* and **21 kg·m/s** *south*. Its combined mass is **3 kg**. **Find final velocity components**.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: shift = Δv × time; J = MΔv.
Fictional calculation that treats the motion as a straight line. M is target mass; J is required impulse. This is not a real encounter forecast.

Worked reasoning: A **1 kg** cart at **+6 m/s** sticks to a **2 kg** cart at rest. Final velocity is **+2 m/s**. The moving cart changes momentum by **-4** and the other by **+4 kg·m/s**.

Your turn: Draw both momentum changes and **show how their sum matches** the system change.

Sentence starter: In my work, I used … to show … . I labeled it … .

Before	After

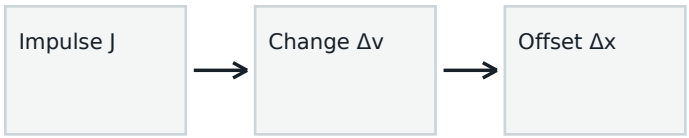
Label mass, velocity and momentum.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Organize this chain with units: impulse → velocity change → sideways shift. Which step uses target mass?

Sentence starter: I notice … . This supports … because … .

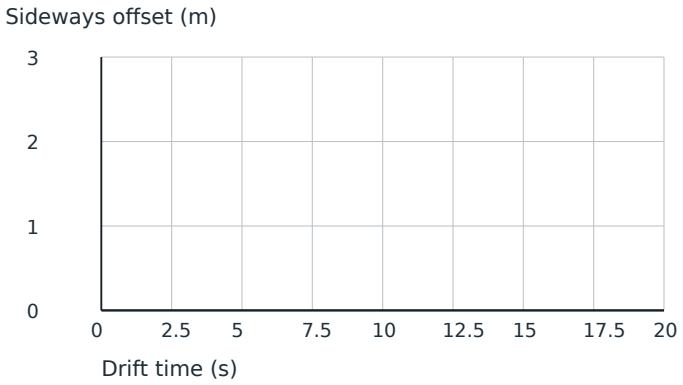


Marzano check - circle: ① ② ③

04 Apply the physics

In a simplified deflection simulation, a **100 kg** target needs a **2 m** sideways offset after a **20 s** drift. Assume the added sideways velocity stays constant after a brief push. **Find the required velocity change**, then the impulse.

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: ① ② ③

Momentum analysis: put it together

antocciphysics

Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

If the target mass estimate is too low, how would a fixed delivered impulse's velocity change compare with the prediction?

Word help: estimate = find an approximate value

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A **4 kg** cart moving at **6 m/s** stops in **3 s**. **Find momentum-change magnitude** and average stopping-force magnitude.
Use before and after snapshots to organize the momentum account.

Word help: momentum = mass times velocity, **including direction**

Sentence starter: I found … [unit]. This value means … in this situation.

Before	After

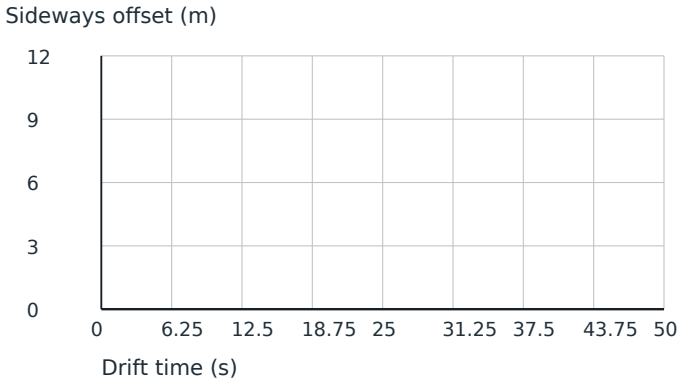
Label mass, velocity and momentum.

Marzano check - circle: 1 2 3

07 Exit check

A brief push adds **0.2 m/s** sideways to a probe's velocity. The probe then drifts for **50 s**. **Complete the position-time line**, find its sideways offset and **state one thing you** are treating as true.

Sentence starter: I think … . I can point to …, which shows …, so … .



Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Return to the worked example in problem **2**. The same pair now receives an external impulse of **3 N·s** right during contact. **Find final total momentum** and shared velocity. Organize the known quantities on the before-and-after diagram.

Sentence starter: I found … [unit]. This value means … in this situation.

Before	After

Label objects and signed velocities.

Marzano check - circle: 1 2 3